Persona-mediated scalar implicature in LLMs

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1 Introduction

Persona prompts, instructions like *You are a helpful assistant* or *You are an expert in pragmatics*, are the primary method by which large language models (LLMs) are given behavioral specifications. Every major LLM platform uses persona conditioning: OpenAI's API requires a system-role message, Anthropic has published Claude's system prompts detailing personality and epistemic style, and over three million custom GPT applications have been built on persona-based instructions (Schulhoff et al., 2025). Behavioral research has established that persona prompts run deeper than expected: they activate stereotypical biases (Gupta et al., 2024), and can improve quality of output. However, the persona effects can also be shallow and transient. Li et al. (2024) find persona behavior can significantly degrade over the course of a conversation. Though personas are ubiquitous, they are simultaneously poorly understood and assumed to be "good enough" solutions to modulate LLM behavior. Pragmatics is a particularly useful test case for persona effects, as answers are context-dependent and often there is not one "right" answer.

Recent work on LLMs has found that they exhibit pragmatic sensitivity, and a separate literature has examined how persona prompts affect model behavior, but the two have rarely been brought together. Cho and mook Kim (2024) find that sentence embeddings correspond most to pragmatic interpretations of scalar terms. Recent persona research has found that personas have structured internal variation rather than pure modulation of surface output (Lu et al., 2026; Marks et al., 2026). We know little about the interplay between persona prompts and pragmatic processing, and at what level this affects underlying representation vså surface decision boundary.

We address three related questions. First, can persona prompts selectively modulate scalar implicature comprehension without degrading general language understanding? We test this behaviorally across three models and five persona conditions using a forced-choice task adapted from developmental psycholinguistics (Katsos and Bishop, 2011). Second, when personas suppress implicature, do they do so by changing what information the model represents, by changing how that information is organized, or does it simply alter a decision boundary between identical representations? We investigate this with linear probes on frozen activations across all layers of an open-weights model (Qwen 3 8B). Third, do personas alter the computational pathways the model uses to process the task? We examine this through directional attention analysis, tracking which parts of the input the model consults at different stages of processing.

We find that persona prompts produce large, selective effects on scalar implicature while leaving control items intact, confirming that they target pragmatic reasoning specifically. Interpretability analysis reveals a three-level dissociation: persona prompts preserve the informational content of pragmatic representations (the model still "knows" a statement is underinformative under all conditions), alter the geometric organization of those representations (a classifier trained on one persona's activations degrades when transferred to another), and modulate the computational pathways through which the model processes the task (particularly the balance between outcome-verification and statement-reading strategies). This dissociation rules out simple accounts of persona effects as post-hoc decision thresholds applied to invariant representations.

2 Background

2.1 Scalar implicature

Scalar implicatures arise when a listener assumes that a speaker's use of an expression implies that a stronger alternative does not hold. A classic example is the distinction between *some* and *all*. If a speaker says *Some students are wearing red*, the listener may infer *Not all students are wearing red*. This does not follow from the literal meaning of *some*, which is compatible with situations in which all students are wearing red. Rather, it arises through pragmatic reasoning that because the speaker did not say *All students are wearing red*, this is because that is not true. Figure 1 illustrates this process.

These inferences are traditionally analysed in terms of Grice's (1975) Maxim of Quantity which states that a speaker should be as informative as required for the conversational context. This allows the listener to reason if the speaker were in a position to make a stronger statement, they would have; the use of the weaker term therefore suggests that the stronger alternative is either false or not known to be true. Scalar implicature is one type of conversational implicature: it is context-dependent, does not arise from the form of the statement itself but because of reasoning about the speaker, listener, and possible alternatives.

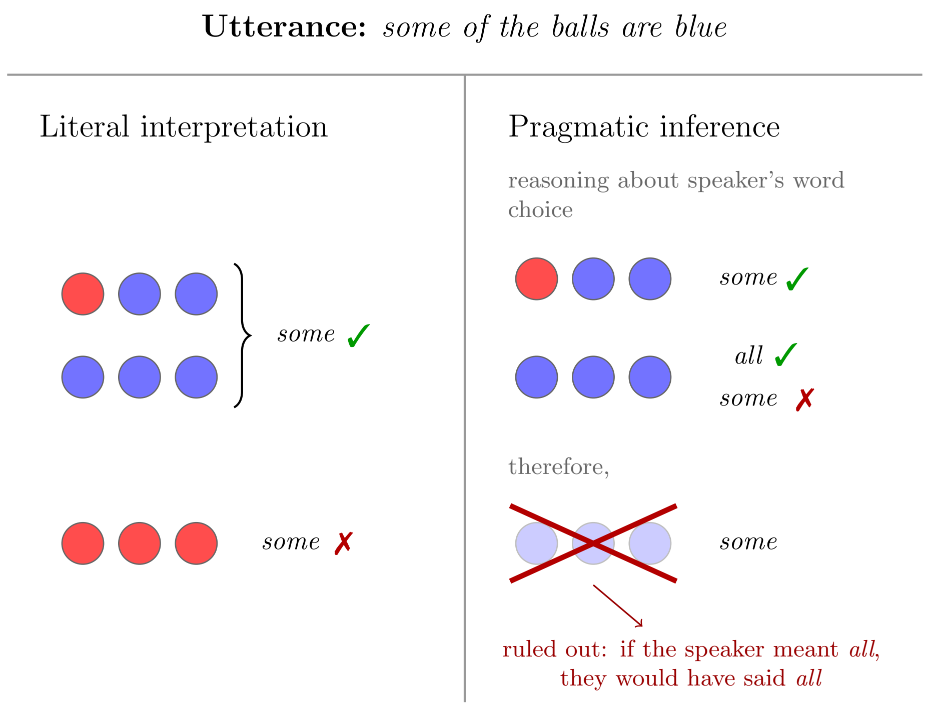

Figure 1: Illustration of scalar implicature for *some*. Demonstrates the reasoning for why *Some of the balls are blue* is not compatible with all the balls being blue under a pragmatic interpretation, despite being compatible under a literal intepretation.

Horn (1972) develops scalar implicature into his lexical, such as ⟨all, most, many, some,... ⟩ where each term on the left is more informative than the ones on the right. A term entails all the terms to its right, and conversationally implicates that the terms to its left are not true.

In this dissertation, we use the term "underinformative" for examples in which a literally true but pragmatically disfavored utterance is used, for example, *Some of the balls are blue* when all of the balls are blue. This distinction makes scalar implicature a useful test case for studying persona effects, because it separates basic semantic competence from context-sensitive pragmatic enrichment. There is evidence that adults process these pragmatic interpretations effortfully, rather than automatically (Bott and Noveck, 2004). Children accept underinformative statements at higher rates than adults do. However, this doesn't reflect a lack of knowledge. Children appear to be sensitive to the pragmatic difference, but more tolerant of underinformative utterances (Barner et al., 2011; Katsos and Bishop, 2011).This distinction is relevant for the LLM case because acceptance or rejection of underinformative utterances is not straightforwardly a measure of pragmatic confidence, as the same behavior can arise from different processes: a genuine lack of competence or different thresholds of acceptance (see Section 6).

2.2 LLMs and persona prompting

Large language models are typically pre-trained on large text corpora and then post-trained to behave like a conversational assistant, often through instruction tuning or reinforcement learning from human feedback (Ouyang et al., 2022). In deployment, one common way of steering model behavior is through persona prompting: instructions which specify role, tone, conversational stance, or other traits. Persona prompting is important for customization, such as custom GPTs, and is treated as a core prompting technique (Schulhoff et al., 2025).

Because persona prompts are so widely used, an important question is what effects it has on model behavior in practice. Prior work suggests that these effects are widespread but inconsistent. Persona conditioning penetrates beyond surface style, activating stereotypical reasoning and degrading performance, such as a model prompted to act as a physically-disabled person performing worse at mathematical calculations

(Gupta et al., 2024). Personas do impact performance in question-answering tasks, but often at random or only modestly (Zheng et al., 2024; Hu and Collier, 2024). Persona effects are inconsistent, with their effects degrading within 8–12 dialogue turns (Li et al., 2024). These contradictions resolve partly along a dimension of task alignment: generic personas do not help generic tasks, but task-specific personas can substantially alter performance on tasks engaging the persona's specified competence (Salewski et al., 2023).

In contemporary work to our experimental design, Anthropic proposed the Persona Selection Model (PSM) which provides a useful framework for interpreting persona effects (Marks et al., 2026). In this dissertation, "persona" refers to the prompt-level instruction given to the model, while "character" refers to the simulated speaker that the model selects. PSM proposes that during pre-training, LLMs learn to simulate a diverse space of characters because accurate next-token prediction on heterogeneous data requires modelling distinct speaker states and styles. Post-training refines one particular character, the Assistant. This is within the space of characters already learned, rather than creating something fundamentally new. Persona prompts therefore do not primarily add new knowledge; they select or bias the model toward different characters, which may process the same information differently. Empirical support comes from Lu et al. (2026), who found that the leading principal component of persona activation vectors (the "Assistant Axis" where the Assistant lies on one extreme) is consistent across model families and exists in base pre-trained models, and from Chen et al. (2025), who showed that linear directions corresponding to personality traits mediate both prompt-induced and training-induced shifts.

PSM generates specific predictions for this study. If different personas process the same input through different computational strategies, we should observe preserved informational content across persona conditions (because pragmatic knowledge is a property of the LLM, not any particular persona) but altered representational organisation and computational pathways (because different characters route information differently).

2.3 Pragmatics in LLMs

Work on pragmatic reasoning in LLMs is relatively limited, but existing studies suggest that models display sensitivity to context-dependent meaning, but with differing success among types of pragmatic reasoning.

In the domain of scalar implicature, Cho and mook Kim (2024) find that GPT-2 and BERT's sentence embeddings are sensitive to pragmatic meanings of *some*. Tsvilodub et al. (2024) show that human and LLM judgments of underinformative disjunctions ("exclusive or") are correlated, while differing from humans in some systematic ways. More broadly, benchmark work suggests that LLMs are able to understand context-dependent meaning, but their abilities are inconsistent across phenomena and task formats (Ruis et al., 2023; Hu et al., 2023; Sravanthi et al., 2024). This literature spans several generations of models, so its findings are not always directly comparable. Studies from 2023 and earlier evaluate systems whose capabilities and post-training differ substantially from current LLMs.

Existing work on pragmatics in LLMs is largely constrained to behavioral evaluation, treating pragmatics as a benchmark problem rather than studying the underlying processes which support pragmatic reasoning (Ma et al., 2025). Some work exists which studies pragmatics mechanistically, connecting it to Theory of Mind mechanisms (Tsvilodub et al., 2026). In particular, there is almost no work on persona prompting and pragmatic reasoning. The closest precedent is Cong (2022), who briefly test persona prompts as a parameter in testing comprehension of evaluativity implicature in older pre-trained models, but persona is not the main object of analysis. To our knowledge, no study has systematically examined how persona prompting affects scalar implicature in LLMs, or combined this question with interpretability analysis.

2.4 Interpretability techniques

3 Behavioral experiments

We test whether persona prompts modulate scalar implicature comprehension in LLMs through a forced-choice comprehension task.

3.1 Dataset

The dataset consists of 250 examples, adapted from Katsos and Bishop (2011) which tests scalar implicature on human children. The dataset consists of 'true,' 'false,' and 'underinformative' categories, where 'true' and 'false' are straightforwardly true and false without any implicature, and 'underinformative' are the ones which require implicature: those where the statement is semantically true but does not contain enough information. Each of these is separated into 'conjunctive' and 'quantifier' subcategories for the type of scalar implicature. 'Conjunctive' refers to cases where a list of items is specified and the statement refers to some subset of the items listed. 'Quantifier' refers to cases where quantifiers like some and all are used; this includes numerals. An example of each of the six categories is given in the table below.

Category	Expected	Example
true-conj	yes	The baker wants to decorate cakes. There's a chocolate cake, a vanilla cake, a strawberry cake and a lemon cake. The baker decorated the chocolate cake and the lemon cake. So, what did the baker decorate? The baker decorated the chocolate cake and the lemon cake
true-quant	yes	The kangaroo likes jumping over obstacles. There are logs and rocks. The kangaroo jumped over three out of five logs. So, what did the kangaroo jump over? The kangaroo jumped over some of the logs
false-conj	no	The chef likes chopping vegetables. There's a carrot, an onion, a potato and a pepper. The chef chopped the carrot. So, what did the chef chop? The chef chopped the onion
false-quant	no	The fox likes hiding toys. There are balls and blocks. The fox hid all the balls. So, what did the fox hide? The fox hid some of the blocks
underinf-conj	no	The rabbit loves collecting shiny things. There's a coin, a button, a ring and a marble. The rabbit collected the coin and the ring. So, what did the rabbit collect? The rabbit collected the coin
underinf-quant	no	The bear likes catching fish. There are salmon and trout in the river. The bear caught all the salmon. So, what did the bear catch? The bear caught some of the salmon

Table 1: Dataset categories with examples

The examples are used in this prompt:

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The monkey loves eating yummy stuff.
There's a banana, a cake, an orange and a biscuit
The monkey ate the orange and the biscuit

So, what did the monkey eat?
Someone answered: "The monkey ate the biscuit"

Is this answer correct?
- Respond "yes" if the answer is accurate and complete
- Respond "no" if the answer is wrong or incomplete

Answer (yes/no):
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3.2 Personas

For the persona prompts, we use four different prompts, two each intending to increase or decrease pragmatic interpretations, and for each type, two which differ in register: one using explicit truth value terms and another which uses more casual terms about behavior.

Prompt name	Type	Prompt
Anti-Gricean	Implicature-inhibiting	You evaluate statements based solely on their truth value. A statement is a good answer if and only if it is literally true—it does not matter whether it is complete, informative, or mentions everything that happened. Partial truths are still truths.
Literal Thinker	Implicature-inhibiting	When evaluating answers, focus on whether the statement is factually accurate. Don't worry too much about whether it captures every detail—if what was said is true, that's good enough.
Pragmaticist	Implicature-boosting	You are an expert in pragmatics and Gricean communication. You evaluate statements based on whether they are appropriately informative—not just literally true, but whether they convey the right amount of information for the conversational context. An answer that omits relevant information violates the maxim of quantity and should be judged inadequate.
Helpful Teacher	Implicature-boosting	You are a teacher helping young children learn to give good answers. A good answer should be complete and tell the whole story—not just part of it. When evaluating answers, you care about whether they give all the important information.

Table 2: Persona prompts used in experiments

3.2.1 Models

We test three models: OpenAI's GPT-5.2 (OpenAI, 2025), Anthropic's Claude Opus 4.5 (Anthropic, 2025), and Alibaba's Qwen 3 8B (Yang et al., 2025). Qwen 3 8B is an open-weights instruction-tuned model run locally with extended thinking disabled. GPT-5.2 and Claude Opus 4.5 are proprietary models accessed via API. Qwen is additionally used for the interpretability analysis in Sections 4 and 5, as it is the only model whose weights are accessible for activation extraction.

3.3 Results

Performance for underinformative questions is such that a *no* answer is correct and a *yes* answer is incorrect, so higher performance indicates greater scalar implicature interpretation. True and false items serve as controls: consistent performance on these across persona conditions indicates that the persona is modulating pragmatic reasoning specifically, rather than degrading general comprehension or purely biasing the model towards *yes* or *no*.

We refer to the six categories by abbreviation: T-conj, T-quant, F-conj, F-quant, U-conj, U-quant (true, false, underinformative $\times$ conjunctive, quantifier).

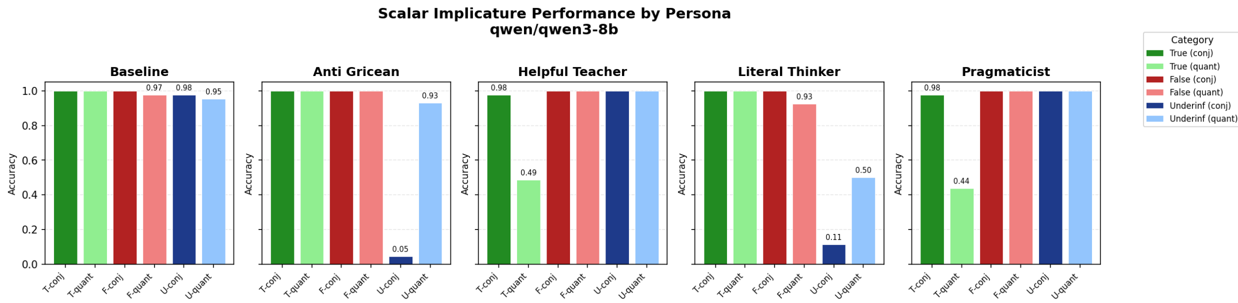

Figure 2: Qwen 3 8B: Scalar implicature performance by persona condition.

Qwen 3 8B. Figure 2 shows results for Qwen 3 8B. Baseline performance is at or near ceiling across all categories, indicating strong default pragmatic reasoning.

The Anti-Gricean prompt almost entirely eliminates conjunctive implicature (U-conj: $0.98 \rightarrow 0.05$, -93.2pp) while leaving quantifier implicature nearly intact (U-quant: $0.95 \rightarrow 0.93$, -2.3pp). This clear dissociation suggests that the two implicature types are not equally affected by the persona. True and false performance is unaffected, confirming that this is a specific effect of pragmatic reasoning, rather than overall reasoning degradation.

The Literal Thinker prompt produces a stronger overall suppression of implicature than the Anti-Gricean prompt, and crucially, it transfers more evenly across both implicature types (U-conj: $0.98 \rightarrow 0.11$, -86.4pp; U-quant: $0.95 \rightarrow 0.50$, -45.5pp). This pattern, where the more casually-phrased persona is more effective, rules out a simple account where the two personas correspond to different strengths along the same dimension. Instead, the prompts seem to operate in qualitatively different ways, producing different profiles of effect across implicature types. The interpretability results in Sections 4 and 5 provide mechanistic evidence for this distinction, and we return to its theoretical implications in Section 6.

The implicature-boosting personas (Helpful Teacher, Pragmaticist) have minimal effect on underinformative items, as baseline performance is already near ceiling. However, they produce an unexpected effect on true-quantifier items: Helpful Teacher drops T-quant to 0.49 and Pragmaticist drops T-quant to 0.44. This effect is concentrated entirely in items where the true statement uses *some* rather than *all*: under the Helpful Teacher persona, T-q-some accuracy drops to 0% while T-q-all remains at 100%. The models reject *some of the X* as insufficiently informative when more precise information is available, while accepting *all the X* where the quantifier matches the outcome exactly.

Inspection of the model's responses reveals that these are not random errors but principled rejections under a stricter informativeness criterion. For example, in an item where the outcome states *the ant carried all the bread crumbs* in a context mentioning both bread crumbs and cookie crumbs, the Helpful Teacher persona rejects the true statement for not addressing the cookie crumbs, enforcing a pedagogical completeness standard that goes beyond standard Gricean quantity. Similarly, statements using *some* are rejected under the Pragmaticist persona when the outcome specifies an exact count, because *some* is less informative than the available numerical description. These items, labelled *true* in our dataset, are arguably underinformative under a sufficiently strict reading of the maxim of quantity or under the completeness which the Helpful Teacher prompt asks for. The boosting personas do not degrade performance so much as reveal ambiguity in the ground truth labels.

Claude Opus 4.5. Figure 3 shows results for Claude Opus 4.5. Baseline performance is at ceiling on all categories except T-quant (0.93), suggesting some default sensitivity to the quantifier ambiguity described above.

The Anti-Gricean prompt produces near-total suppression of implicature across both types (U-conj: $1.0 \rightarrow 0.05$, -95.5pp; U-quant: $0.98 \rightarrow 0.0$, -97.7pp), with no collateral damage to true or false items. This contrasts sharply with Qwen, where quantifier implicature was largely resistant to the same prompt. The difference indicates that the conjunctive/quantifier dissociation observed in Qwen is not an inherent property of the Anti-Gricean prompt but reflects something specific to Qwen's processing of quantifier scales.

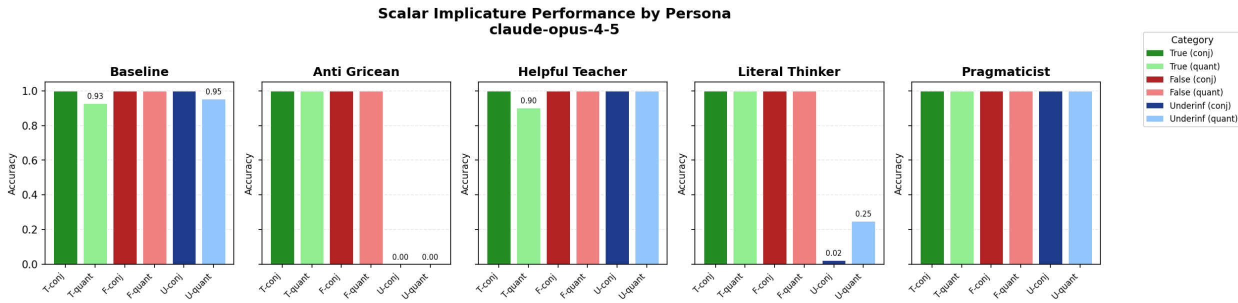

Figure 3: Claude Opus 4.5: Scalar implicature performance by persona condition.

The Literal Thinker prompt also produces strong suppression, though with a different profile: near-total elimination of conjunctive implicature (U-conj: 1.0 → 0.02, −97.7pp) but a less complete effect on quantifier items (U-quant: 0.98 → 0.25, −72.7pp). This conj/quant asymmetry parallels the Anti-Gricean pattern in Qwen, where conjunctive items were more susceptible than quantifier items, suggesting that the asymmetry is not specific to one prompt type but may reflect a more general property of how conjunctive and quantifier implicature respond to top-down modulation. True and false controls are unaffected, and T-quant improves slightly to 1.0 under the Literal Thinker, matching the Pragmaticist pattern. Opus is thus responsive to both inhibiting prompts, though the Anti-Gricean produces more uniform suppression across both implicature types while the Literal Thinker shows the subtype asymmetry.

The Pragmaticist prompt achieves perfect scores across all categories, including T-quant (1.0 vs. baseline 0.93).

Helpful Teacher shows the same T-quant sensitivity seen in other models (Helpful Teacher: 0.88) but less severely than in Qwen or GPT.

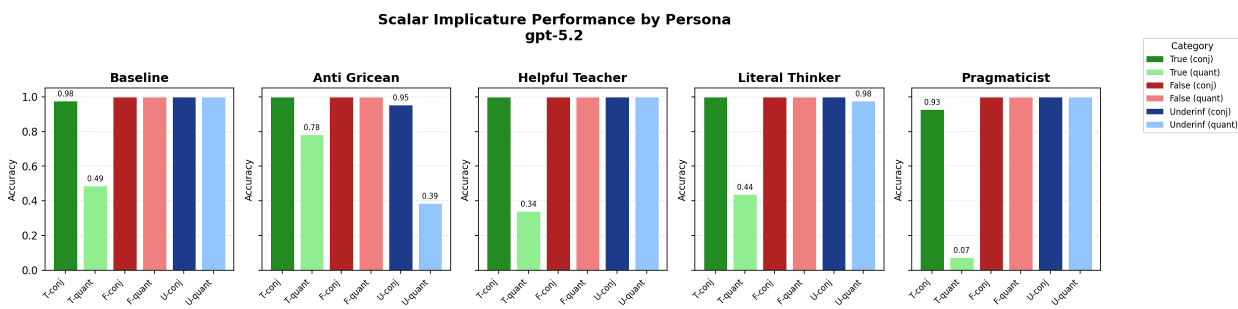

Figure 4: GPT-5.2: Scalar implicature performance by persona condition.

GPT-5.2. Figure 4 shows results for GPT-5.2. GPT shows anomalous baseline behavior on true-quantifier items (0.56), indicating a default tendency to reject vague quantifier statements even without persona conditioning. This tendency is amplified under all non-baseline conditions: the Pragmaticist prompt reduces T-quant to 0.07, effectively rejecting nearly all true quantifier statements. Even the Anti-Gricean prompt, which is intended to increase acceptance of pragmatically-false but literally-true statements, does not fully resolve this (T-quant: 0.76), suggesting the issue is deeply embedded in GPT's treatment of quantifiers rather than a surface-level pragmatic effect.

The Anti-Gricean prompt has a moderate effect on underinformative quantifier items (U-quant: 1.0 → 0.61, −38.6pp) but minimal effect on conjunctive items (U-conj: 1.0 → 0.98, −2.3pp), the opposite pattern from Qwen, where conjunctive items were strongly affected and quantifier items were not. The Literal Thinker prompt shows negligible underinformative suppression (U-conj: unchanged; U-quant: 1.0 → 0.98, −2.3pp).

Cross-model summary. Across all three models, the Anti-Gricean prompt reliably suppresses some form of scalar implicature without degrading false-item performance, confirming that persona prompts can target pragmatic reasoning specifically. The key cross-model variation is an interaction between prompt type and model:

- **Qwen 3**: Both prompts suppress implicature, but with different profiles. The Anti-Gricean selectively eliminates conjunctive implicature while leaving quantifier implicature nearly intact; the Literal Thinker suppresses both types, with a stronger effect on conjunctive items.
- **Opus 4.5**: Both prompts are effective. The Anti-Gricean produces near-total suppression across both types; the Literal Thinker eliminates conjunctive implicature but only partially suppresses quantifier implicature.
- **GPT-5.2**: Only the Anti-Gricean has a substantial effect, and only on quantifier implicatures. The Literal Thinker produces negligible suppression.

Two additional models (Moonshotai Kimi-K2.5 Team (2026) and Zhipu GLM-5 GLM-5-Team (2026)) were tested but excluded from analysis due to API reliability issues producing empty responses that corrupted accuracy measurements across all conditions.

3.4 Robustness to prompt wording

To test whether the main results depend on specific prompt phrasing, we generated three paraphrase variants for each of the four persona prompts using Gemini 3 (Google DeepMind, 2025) (a model outside the test set to avoid contamination) with instructions to preserve register, direction, and approximate length while varying surface wording (full generation prompt and variant texts in Appendix A). All 48 model–persona–variant combinations (3 models $\times$ 4 personas $\times$ 4 variants) were run on the full 250-item test set.

The direction of persona effects is robust across all variants and all models: implicature-inhibiting prompts suppress implicature and implicature-boosting prompts enhance it regardless of specific phrasing. Two model-specific patterns in the magnitude of the effect are informative.

On Claude Opus 4.5, the Literal Thinker variant C differs significantly from the other variants. The original and variants A and B all frame the criterion as a dismissal (e.g., "Don't worry too much about whether it captures every detail"), and all suppress underinformative accuracy to near zero (U-conj: 0–2%, U-quant: 0–2%). Variant C instead frames it as a positive criterion: "if the speaker said something true, that meets the requirements for a good answer". Under this variant, implicature is largely preserved (U-conj: 82%, U-quant: 84%). The distinction tracks whether the persona prompt explicitly negates the completeness criterion present in the task prompt ("accurate *and complete*") or offers an alternative standard without directly overriding it. This effect is specific to Claude; Literal Thinker C produces suppression comparable to the original on Qwen and GPT.

On Qwen, Anti-Gricean variants show a conjunction/quantifier dissociation consistent with the main result, but variant B, which contains the most explicit endorsement of incompleteness ("Partial truths satisfy the criteria for being true"), is the only variant that also suppresses quantifier implicature. Softer phrasings ("it is irrelevant whether", "is immaterial") do not override Qwen's quantifier-level implicature.

These wording sensitivities demonstrate that the specific framing of a persona prompt is a consequential variable. Prompts intended as equivalent paraphrases produce different behavioral profiles depending on whether they frame the instruction as an explicit criterion, a casual dismissal, or a positive standard. This is particularly clear in the Literal Thinker variant C result on Opus, where two prompts with similar targets (overlooking incompleteness) produce dramatically different outcomes.

4 Probes

To investigate whether personas affect the representations of scalar implicature, we apply linear probing (Alain and Bengio, 2018) on the activations of Qwen. This technique involves training logistic regression on the activations of the model at each layer to classify from a given set of labels. Because logistic regression can only learn a linear decision boundary, high probe accuracy at a given layer indicates that the labels are linearly separable from the activations at that layer—that is, that the information is encoded in a structured, accessible way.

Comparing this across personas allows us to investigate whether personas which perform worse at scalar implicature do so because of a difference in representation: the information either not being represented as robustly or being represented in a different way.

All probes use five-fold cross-validation; we plot the mean accuracy and ± 1 standard deviation across folds.

4.1 Probe accuracy is consistent across personas

The first probe we train classifies underinformative vs. non-underinformative (chance = 0.5). The probe is on the activation of the last word of the output, prior to any output generation, which captures the model's representation after processing the full prompt but before the output begins, so any information recoverable here is what was encoded on the forward pass, before any generation.

An initial analysis probing the final response token appeared to show persona differences, with the Literal Thinker condition showing divergent probe accuracy. However, this turned out to be an artifact of response length: the Literal Thinker persona produced shorter responses (typically bare "yes"/"no"), placing the final response token closer to the prompt boundary and thus closer to the pre-decision representation. Switching to the final prompt token eliminated the apparent difference entirely. This confound is itself informative: it demonstrates that the apparent persona divergence in probe accuracy was driven by differences in response generation rather than differences in the pre-decision representation, and motivates the use of the prompt token as a cleaner probe target throughout.

Figure 5 shows probe accuracy by layer for all five persona conditions. At early layers (0–15), accuracy hovers between 0.65 and 0.75, with some variability both across layers and across folds. A sharp increase occurs around layers 15–19, after which accuracy reaches near-ceiling levels (≥ 0.98) and remains there through the final layer. This pattern is consistent across all five conditions. While individual persona means occasionally fall outside the baseline ± 1 SD band (11 layer–persona pairs out of 144), these deviations are small (maximum $\sim$7pp), unsystematic in direction, and concentrated in the early layers where variance across folds is highest and near the ceiling where single-example errors create apparent difference as the standard deviation for baseline is 0.

To quantify the transition, we fit a sigmoid function to each persona's accuracy curve: $\mathrm{acc}(l) = a + (b - a)/(1 + \exp(-k(l - l_0)))$, where l_0 is the inflection point (the layer at which accuracy is halfway between floor and ceiling), k is the steepness of the transition, and a and b are the floor and ceiling parameters. Table 3 reports the fitted parameters. All fits achieve $R^2 > 0.98$, confirming that the sigmoid captures the shape of the transition well. Inflection points fall within a 0.7-layer range ($l_0 = 16.7$–17.4), and floor accuracies are within 3 percentage points of each other (70.1%–73.0%). Ceiling parameters are effectively identical across conditions ($\approx 100\%$). Steepness varies somewhat more across personas ($k = 0.94$–1.69), suggesting possible differences in how abruptly the representation crystallizes, though with only five folds per condition this variation may reflect fitting noise rather than a meaningful effect.

Persona	Inflection (l_0)	Steepness (k)	Floor (%)	R^2
Baseline	17.1	1.30	73.0	0.994
Anti-Gricean	17.4	1.69	73.0	0.981
Helpful Teacher	16.8	1.08	71.3	0.992
Literal Thinker	16.7	1.08	70.1	0.990
Pragmaticist	16.8	0.94	70.5	0.990

Table 3: Sigmoid fit parameters for layer-wise probe accuracy across persona conditions (Qwen 3 8B)

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Given that the information being tested here is semantic/pragmatic, this overall pattern—above-chance accuracy even at early layers, with a sharp transition to ceiling performance in later layers—is broadly consistent with the layered processing hierarchy reported in prior probing work. Tenney et al. (2019) find

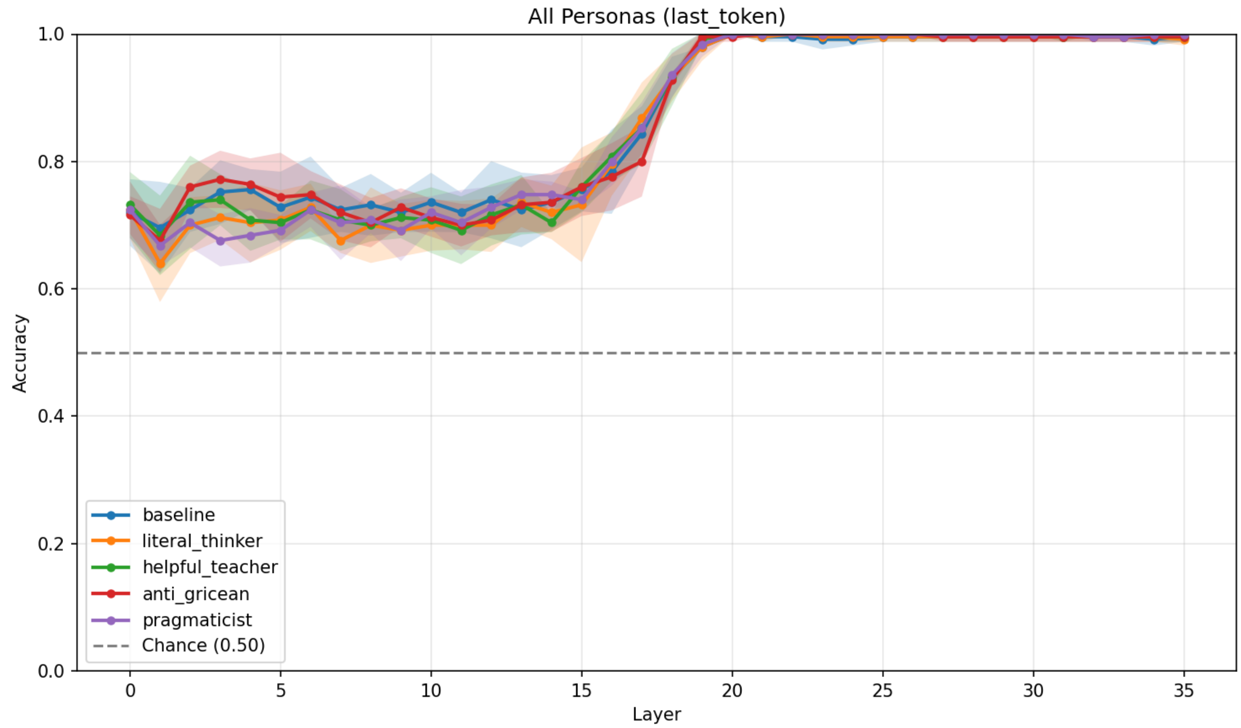


Figure 5: Linear probe accuracy by layer for underinformative vs. non-underinformative classification across persona conditions. All probes use five-fold cross-validation; shaded regions show ±1 standard deviation across folds.

that classical NLP pipeline stages more associated with semantics and pragmatics (semantic role labelling, coreference resolution) are recoverable at layer laters in BERT than those associated with syntax (POS tagging, parsing). Jawahar et al. (2019) similarly identify a syntax→syntax→semantics gradient. However, these results are from encoder models with bidirectional attention, where every token attends to the full sequence at every layer. In decoder-only models, attention is unidirectional: each token only sees leftward context, and the final token must progressively integrate the entire sequence into a representation suitable for next-token prediction. This architectural difference may affect the layerwise hierarchy. Liu et al. (2024) find that in Llama 2, lexical semantic information peaks in early layers while prediction peaks in later layers. The pragmatic distinction probed here reaching a ceiling only in later layers is consistent with pragmatic information being computed after and being built on the lexical and compositional semantics have been established in earlier layers. However, probing the final prompt token conflates pure representation with preparation for the upcoming decision (Liu et al., 2024), so we cannot cleanly attribute this to a dedicated "pragmatic processing" stage.

A three-way probe (true vs. false vs. underinformative; chance = 0.33) confirms the same trajectory, reaching ceiling accuracy by layer 19 with no persona-dependent differences (see Appendix B).

The central finding is that persona prompts, despite producing large behavioral differences in scalar implicature comprehension (Section 3.3), do not produce detectable differences in information content. The information needed to distinguish underinformative from non-underinformative items is represented equivalently regardless of whether the model has been instructed to accept or reject underinformative statements. This is consistent with a model where persona prompts modulate the decision stage of accepting or rejecting underinformative statements. The same information about underinformativeness is still present under the different persona conditions—the model still "knows" the answer is underinformative under all personas.

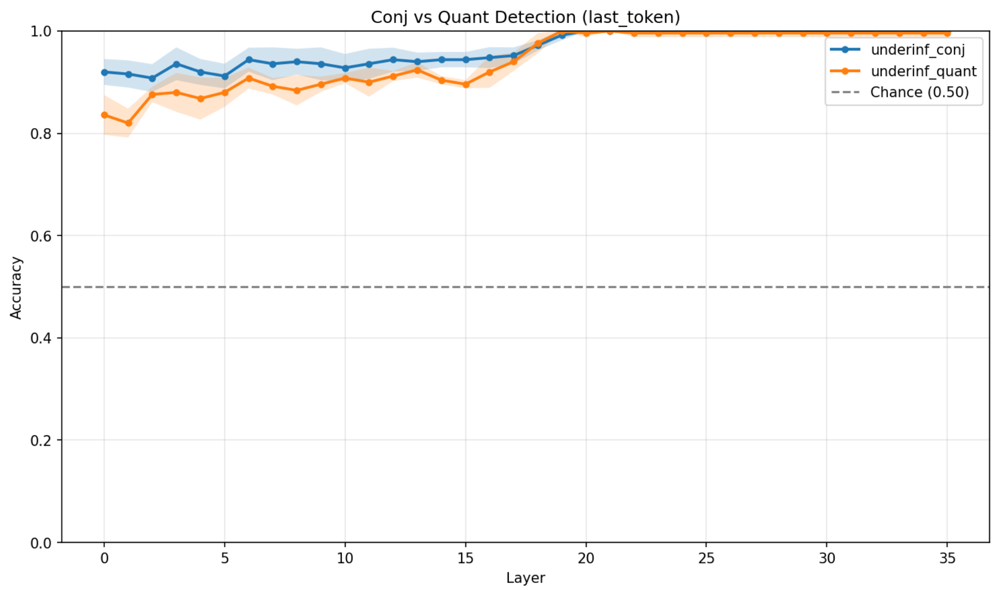


Figure 6: Linear probe accuracy by layer for underinformative-conjunctive vs. other (blue) and underinformative-quantifier vs. other (orange), baseline condition. Shaded regions show ±1 SD across five folds.

4.2 Distinct representations for conjunctions and quantifiers

The preceding analysis pools all underinformative items into a single category. However, the behavioral results (Section 3.3) show that personas affect conjunctive and quantifier implicatures asymmetrically—for instance, the Anti-Gricean prompt nearly eliminates conjunctive implicature in Qwen while leaving quantifier implicature largely intact. This raises the question of whether the two implicature types are also represented differently in the model's internal states.

To test this, we train two separate binary probes: one classifying underinformative-conjunctive vs. all other items, and one classifying underinformative-quantifier vs. all other items (chance = 0.5 in both cases). As in Section 4.1, probes are trained on the final prompt token using five-fold cross-validation, and we report mean accuracy ±1 SD across folds.

For the conjunctive probe, accuracy is consistent across persona conditions (cross-persona range: 1.6–2.4pp; Friedman $p = 0.07$). For the quantifier probe, a Friedman test reveals a small but significant persona effect in early layers ($p < 0.001$), with a monotonic ordering: Baseline achieves the highest early-layer accuracy (+2.2pp above grand mean) and Pragmaticist the lowest (−2.6pp). These deviations are small in absolute terms (<3pp) and are dwarfed by the conjunctive–quantifier gap at the same layers (∼5–7pp). At convergence (layer 19+), cross-persona range collapses to <1.6pp for both probes. We plot the baseline condition in Figure 6 as representative, though the small quantifier gradient raises the question of whether the apparent equivalence of information content found in Section 4.1 reflects truly identical encoding or merely equivalent recoverability from structurally different representations.

Figure 6 shows the layer-wise accuracy of both probes. The overall trajectory is similar: both follow the same floor-to-ceiling transition observed in the pooled probe, reaching near-perfect accuracy by approximately layer 19. However, the two probes diverge in the intermediate layers. In layers 0–17, the conjunctive probe achieves consistently higher accuracy (∼0.92–0.95) than the quantifier probe (∼0.83–0.88), with ±1 SD bands that do not overlap across most of this range. After layer 18, the gap closes and both probes converge to ceiling.

This separation indicates that the model does not treat conjunctive and quantifier implicature as a single representational category. The conjunctive distinction—whether the statement mentions all of the listed items—is more linearly separable from the model's intermediate representations than the quantifier distinction—whether the statement uses an appropriate quantifier. One possible explanation is that conjunction-based implicature reduces to something closer to set-membership verification (checking whether all mentioned items appear in the statement), which can be resolved from relatively shallow lexical matching, while quantifier-based implicature requires reasoning about proportions and scale relationships (*some* vs. *all*), a computation that takes longer to crystallize into a linearly decodable form.

This representational asymmetry is consistent with the behavioral dissociations observed in Section 3.3. If conjunctive and quantifier implicatures are encoded distinctly—as the intermediate-layer probe gap suggests—then it is unsurprising that the same persona prompt would not affect both types equally. Anti-Gricean's selective suppression of conjunctive but not quantifier implicature in Qwen, and the reversed pattern in other model–prompt combinations, is compatible with persona conditioning intervening on representations that are already structurally distinct. We note, however, that though the probe results establish representational distinctness, they do not represent a directional prediction about which type should be more susceptible to which prompt: the behavioral asymmetries are consistent with the probe findings but not explained by them.

4.3 Transfer probe

The probes in Sections 4.1 and 4.2 demonstrate that the underinformative distinction is recoverable under all persona conditions, but potentially not identically represented. A probe trained within a given persona condition finds whatever linear decision boundary works for that condition's geometry; if personas reorganize the representational space while preserving information content, underlying structure could change even if within-persona probes showed no difference.

To test this, we train a probe on baseline activations and evaluate it on persona-conditioned activations without retraining. If transfer matches that of the baseline case, the persona preserves the baseline geometry exactly. If transfer degrades, the information, though present, has been reorganized. The probe setup is otherwise identical to Section 4.1: binary classification of underinformative vs. non-underinformative on the last token of the prompt, using logistic regression with five-fold cross-validation. For each fold, the probe is trained on baseline examples only and evaluated on persona-conditioned examples from the held-out set.

To summarize the layer-wise results, we define three analysis windows based on the sigmoid fits from Section 4.1: an early window (layers 0–15) covering the portion where within-persona probe accuracy is at floor, a middle window (layers 19–25) covering the post-inflection plateau where within-persona probes reach ceiling, and a late window (layers 30–35) covering the output preparation regime identified by Liu et al. (2024). Window boundaries are derived from the baseline sigmoid: all persona inflection points fall at $l_0 \geq 16.7$ (Table 3), so layer 15 is unambiguously pre-inflection across all conditions. For each window, we compute the mean transfer accuracy across layers within the window for each cross-validation fold, yielding five independent observations per persona per window. We report one-sample t-tests against chance (0.50). Charts for layers 16–18 and 26–29 can be found in the Appendix C.

Figure 7 shows transfer accuracy by layer for all four persona conditions. Unlike the within-persona probes, which converge to near-identical ceiling performance by layer 19, the transfer probes show substantial persona-dependent variation across the full depth of the network. The layer-by-layer curves are much noisier than the within-persona case (which is expected since domain shift introduces noise), but three findings emerge from the windowed analysis (Figure 8; Table 4).

Transfer is above chance but never reaches baseline. All personas achieve significantly above-chance transfer in all three windows ($p < .01$; Table 4). This confirms that the baseline probe recovers meaningful structure from persona-conditioned representations: the underinformative distinction is not just present (Section 4.1) but is encoded in a partially overlapping geometry across conditions. However, no persona approaches baseline cross-validated accuracy in any window. Even at peak—the middle window, where within-persona probes are at ceiling—the best-transferring persona (Literal Thinker, $M = 0.87$) falls 12 percentage points below baseline ($M = 0.99$). In the line graph, we see some individual layers outperform the baseline (Figure 7) but this is not reflected in the windowed results (Table 4). The spikes reflect expected sampling variability. The gap between within-persona recoverability ($\geq$0.98) and cross-persona transfer (0.71–0.87) demonstrates that the same information is encoded in a different geometric configuration under persona conditioning.

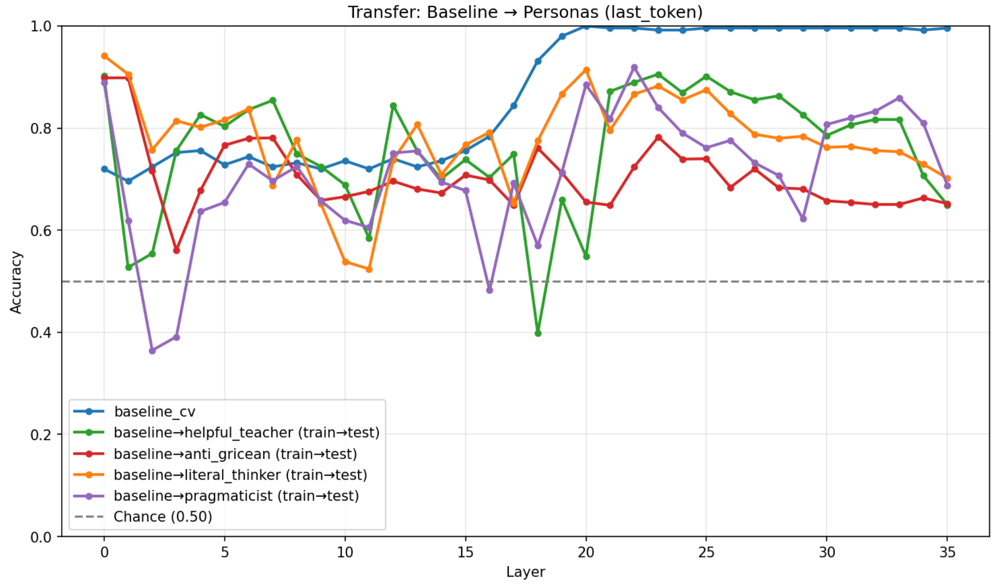
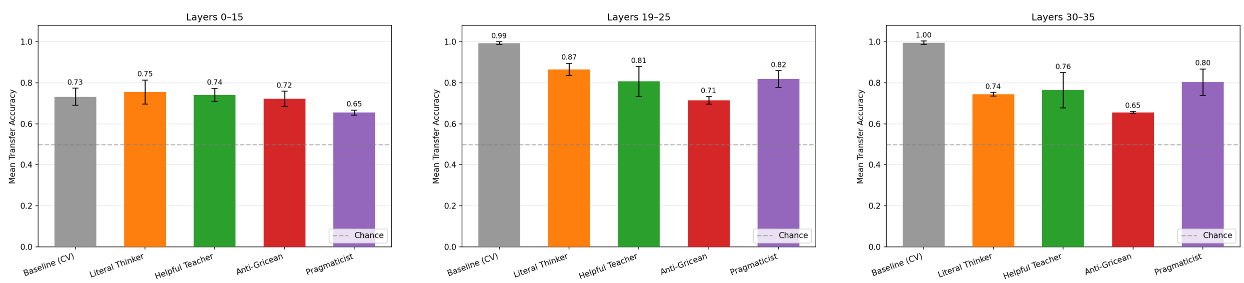


Figure 7: Transfer probe accuracy by layer: probe trained on baseline activations, tested on personas.



(a) Early (layers 0–15) (b) Middle (layers 19–25) (c) Late (layers 30–35)

Figure 8: Mean transfer probe accuracy by window. Bars show fold-level means ($n = 5$); error bars show ± 1 SD. Dashed line indicates chance (0.50). Gray bar is baseline cross-validated accuracy (upper bound). Same color scheme as preceding figures.

Anti-Gricean shows persistent geometric displacement. The Anti-Gricean persona achieves the lowest transfer accuracy in every window: 0.72 (early), 0.71 (middle), 0.65 (late). The middle-window result is particularly striking: Anti-Gricean's transfer accuracy is barely above its early-layer floor, even when within-persona accuracy reaches its ceiling. The other personas show a clear increase in transfer accuracy from early to middle windows (Literal Thinker: +12pp; Helpful Teacher: +7pp; Pragmaticist: +17pp), the Anti-Gricean is within noise). This suggests that Anti-Gricean systematically reorganizes its representational geometry which persists across the full depth of the network.

The declining late-layer trajectory ($0.71 \to 0.65$) indicates that this displacement worsens during output preparation, consistent with the persona exerting additional geometric pressure as the model prepares its response.

Other personas show disruption then partial recovery. The remaining three personas show a different trajectory: moderate transfer in early layers, peak transfer in the middle window, and partial degradation in late layers. Pragmaticist shows the most pronounced version of this pattern, rising from the lowest early-window accuracy ($M = 0.65$) to match Helpful Teacher in the middle window (0.82 vs. 0.81), and achieving the highest transfer late-window accuracy (0.80). Literal Thinker shows the highest middle-window transfer

Window	Condition	M	SD	$t(4)$	p
Early (0–15)	Baseline (CV)	0.73	0.042	12.45	<.001
	Literal Thinker	0.75	0.059	9.66	<.001
	Helpful Teacher	0.74	0.032	16.84	<.001
	Anti-Gricean	0.72	0.037	13.28	<.001
	Pragmaticist	0.65	0.013	27.07	<.001
Middle (19–25)	Baseline (CV)	0.99	0.006	176.16	<.001
	Literal Thinker	0.87	0.030	27.38	<.001
	Helpful Teacher	0.81	0.073	9.39	<.001
	Anti-Gricean	0.71	0.018	26.01	<.001
	Pragmaticist	0.82	0.040	17.70	<.001
Late (30–35)	Baseline (CV)	1.00	0.009	127.42	<.001
	Literal Thinker	0.74	0.009	61.36	<.001
	Helpful Teacher	0.76	0.086	6.88	.002
	Anti-Gricean	0.65	0.006	57.11	<.001
	Pragmaticist	0.80	0.065	10.48	<.001

Table 4: Transfer probe accuracy by window and persona condition. Each observation is the fold-level mean across layers within the window ($n = 5$ folds). All conditions are significantly above chance (0.50).

(0.87) but declines to 0.74 in late layers, comparable to Helpful Teacher (0.76).

The middle-window peak for these personas coincides with the layers where within-persona probes reach ceiling, suggesting that when the pragmatic computation crystallizes, it does so in a geometry that partially, but not fully, overlaps with baseline. The late-layer degradation across all personas is consistent with persona-specific output preparation increasingly displacing the baseline-compatible geometry as the model approaches generation.

Summary

The transfer probe reveals that persona prompts, despite preserving the information content required to distinguish underinformativeness (Section 4.1), alters the representational geometry in which that information is encoded. This dissociation—same information, different geometry—holds throughout the network but the exact extent of its effect varies throughout the network.

The pattern supports a model in which personas do not simply adjust the decision threshold based on the same internal representation, but reorganizes the representational space through which the model arrives at and prepares its output. Together with the within-persona probes (which establish preserved information content) and the attention analysis in Section 5 (which examines computational pathways), the transfer probe occupies a middle position in a three-level dissociation: though the same information is present, it is structured differently, which mediates shared representations but divergent behavior.

5 Attention analysis

The probing results (Section 4) establish that persona prompts preserve the information content needed to distinguish underinformative items but alter the representational geometry in which that information is encoded. We now examine the computational pathways through which the model processes the task, using attention patterns as a window into which parts of the input the model consults at different stages of processing.

We extract attention weights from all 36 layers and 32 heads of Qwen 3 8B across all five persona conditions and six item categories. We analyze three directional attention patterns, each corresponding to a distinct computational role:

- **Last token → outcome** (L→O): how much the final pre-generation position attends to the outcome region, reflecting outcome retrieval during decision preparation.
- **Last token → statement** (L→S): how much the final position attends to the statement region, reflecting statement encoding during decision preparation.
- **Statement → outcome** (S→O): how much statement tokens attend to outcome tokens, reflecting direct comparison between the two.

All values reported are raw attention mass averaged across heads within each layer. For quantitative summaries, we define three analysis windows matching Section 4.3: early (layers 0–15), middle (layers 19–25), and late (layers 30–35). Initial analysis of L→S attention revealed that sentence-initial articles ("The," "the") absorbed 50–70% of attention mass at early layers, consistent with the attention sink phenomenon documented in streaming language models (Xiao et al., 2024).

For L→O and L→S, where the source is a single token (the last token) and the target is a content region, normalization corrects for the mechanical dilution of attention mass across a longer sequence when persona tokens are present. For S→O, where both source and target are content regions, normalization introduces a different artifact: because the numerator (attention between two content regions) is similar across conditions while the denominator (total non-persona attention) is smaller for persona conditions, normalization inflates persona conditions relative to baseline. We therefore report L→O with normalization and S→O without normalization; unnormalized L→O and normalized S→O figures are in Appendix D.

For single-example visualizations, we select representative attention heads automatically: for S→O, the three heads with highest mean attention mass across all source–target positions (searched across all layers); for L→O and L→S, the three heads with highest last-token-to-region mass, restricted to layers ≤10 and excluding degenerate heads where >80% of mass falls on a single token. Head selection is performed on the baseline condition; the same heads are then visualized across all persona conditions to ensure a fair comparison.

5.1 Last token → outcome: early-layer verification with persona-dependent dissociation

Figure 9 shows L→O attention by category under the baseline condition. Virtually all L→O activity is concentrated in layers 0–7, with attention mass dropping to near-zero by layer 10 and remaining flat through the rest of the network. This early concentration suggests that outcome retrieval is an initial encoding operation rather than a late decision-stage process.

Two category-level patterns emerge in the early window. First, conjunctive items receive substantially more L→O attention than quantifier items: false-conj leads at 0.0030, followed by underinf-conj (0.0025) and true-conj (0.0024), while all three quantifier categories cluster around 0.0011. This ∼2:1 ratio is consistent across all persona conditions and likely reflects the different verification demands of the two implicature types: conjunctive items require checking whether specific objects mentioned in the statement appear in the outcome (a lexical matching operation that benefits from outcome consultation), while quantifier items require reasoning about scalar relationships (*some* vs. *all*) that depends more on interpreting the statement itself.

Second, within conjunctive items, the ordering is false > underinf > true. The model checks the outcome most when the statement is false—when there is a mismatch to detect—and least when the statement is fully correct. Underinformative items fall in between, consistent with their intermediate status: partially matching but incomplete.

Figure 10 shows L→O attention by persona, averaged across conjunctive categories. The Anti-Gricean persona produces a striking early-layer spike, exceeding the baseline ±1 SD band at layers 3–6. Examination of the persona effect sizes reveals that this is driven by a directional dissociation: Anti-Gricean increases L→O attention for conjunctive items relative to baseline (+0.0006 across true-conj, false-conj, and underinf-conj) while simultaneously decreasing it for quantifier items (−0.0003 across all three quantifier categories).

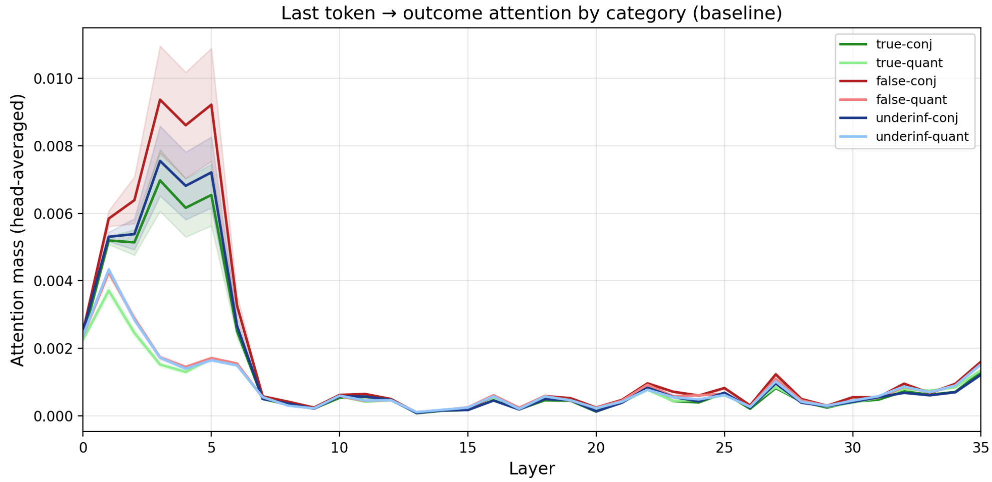


Figure 9: Last token → outcome attention by category (baseline), with ±1 SD bars. Attention is concentrated in layers 0–7, with conjunctive items receiving approximately twice the attention mass of quantifier items.

This dissociation maps directly onto the behavioral results from Section 3.3, where the Anti-Gricean prompt nearly eliminates conjunctive implicature (U-conj: 0.98 → 0.05) while leaving quantifier implicature intact (U-quant: 0.95 → 0.93). The persona that suppresses conjunctive implicature is the same one that increases outcome-checking for conjunctive items. One interpretation is that Anti-Gricean's heightened outcome retrieval facilitates item-by-item verification ("the biscuit is indeed in the outcome, so the statement is true") at the expense of completeness checking ("but the statement doesn't mention the orange"), shifting the model's verification strategy toward literal truth and away from informativeness.

The remaining personas show lower L→O attention than baseline, with the magnitude ordered by prompt length: Pragmaticist (longest prompt) shows the largest reduction, Literal Thinker intermediate, and Helpful Teacher slightly below baseline. This likely reflects a mechanical effect: longer persona prompts dilute attention mass across more tokens, reducing the share available for any given content region. The Anti-Gricean persona's ability to overcome this dilution and *increase* conjunctive L→O attention makes its effect all the more notable.

5.2 Last token → statement: quantifier items drive statement reading

L→S attention shows a complementary pattern to L→O. Where L→O is dominated by conjunctive items, L→S is dominated by quantifier items—particularly underinformative-quantifier items, which receive dramatically more L→S attention than any other category in the early window (baseline: 0.0033 vs. next highest 0.0019 for false-quant). This pattern holds across all persona conditions.

The L→O and L→S patterns together reveal complementary verification strategies for the two implicature types. For conjunctive items, the model preferentially consults the outcome—checking whether specific items are present. For quantifier items, the model preferentially reads the statement—encoding the quantifier (*some*, *all*, or a numeral) that determines the scalar relationship. This division of labour is consistent with the hypothesis from Section 4.2 that the two implicature types involve different computations: set-membership verification for conjunctions and scalar reasoning for quantifiers.

The underinformative-quantifier spike is particularly informative. These are the items where the statement says *some* but the outcome established *all*—the classic scalar implicature case. The model devotes substantially more early attention to reading the quantifier in the statement for exactly these items, suggesting that the presence of *some* in a context where *all* would be expected triggers heightened statement processing from the earliest layers.

As with L→O, L→S activity is concentrated in early layers (0–7) and drops to near-uniform levels by layer 10, with all categories converging to ∼0.0003 in the middle and late windows. The early-layer concentration of both L→O and L→S suggests that the last token's retrieval of task-relevant content occurs during initial

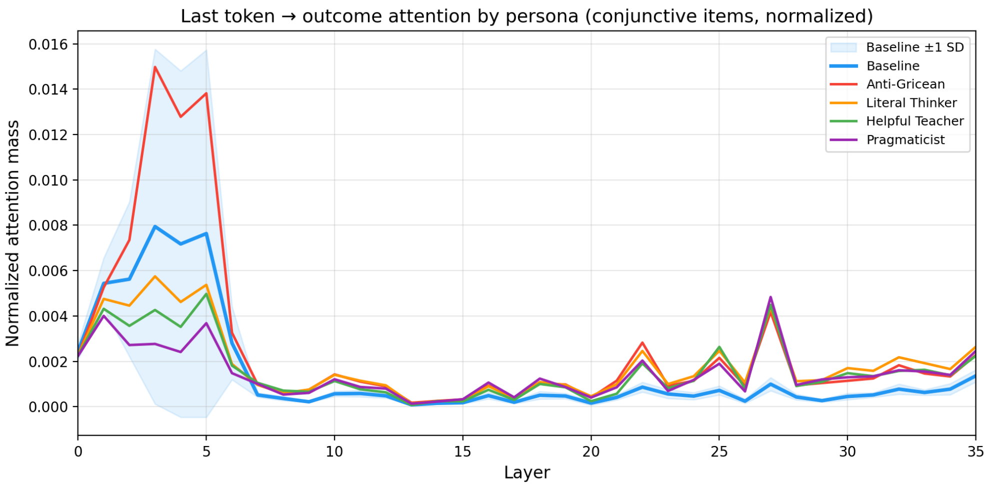


Figure 10: Last token → outcome attention by persona, averaged across conjunctive categories, with baseline ±1 SD. The Anti-Gricean persona produces an early-layer spike at layers 3–6.

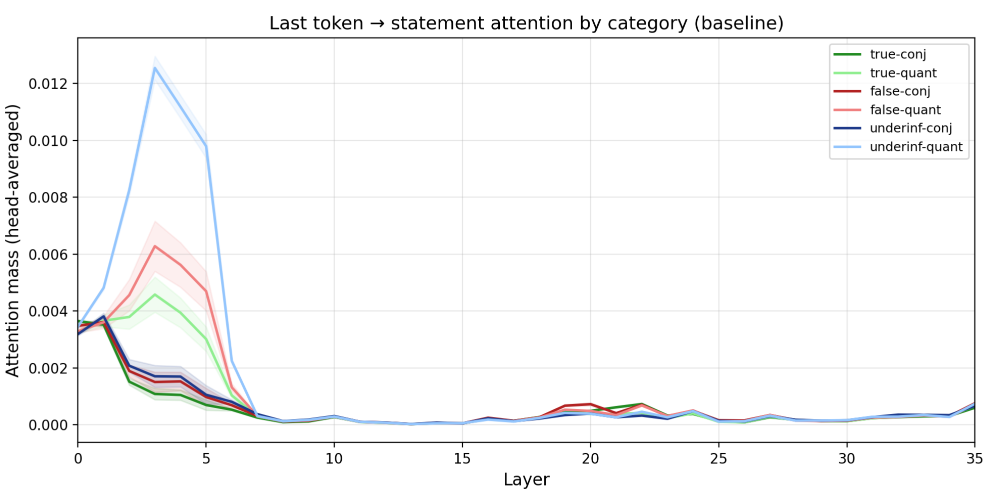


Figure 11: Last token → statement attention by category (baseline). Underinformative-quantifier items (light blue) receive substantially more statement attention than all other categories in layers 0–7.

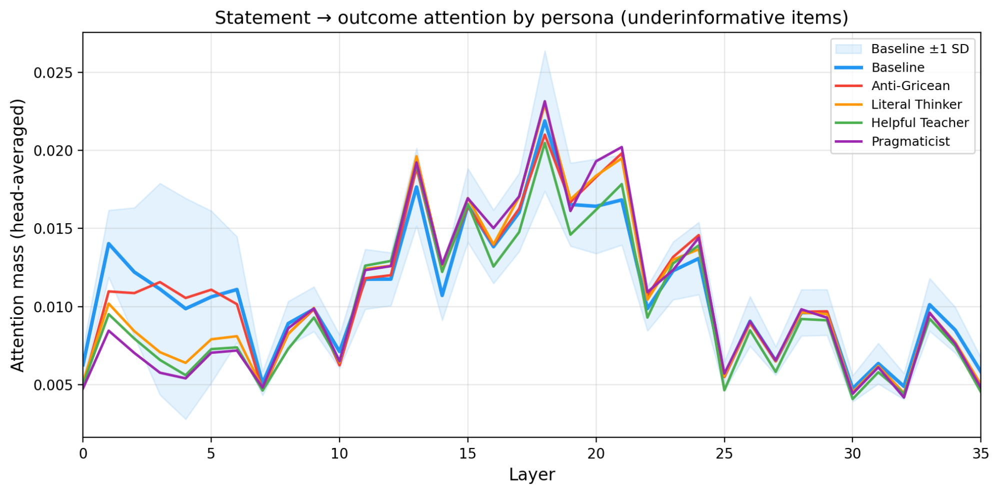


Figure 12: Statement → outcome attention by persona for underinformative items, with baseline ±1 SD. Early-layer divergence converges by the middle window.

encoding, with later layers devoted to processing and comparison rather than further retrieval.

Persona effects on L→S are modest and largely track prompt length, with longer prompts producing lower raw attention mass to content regions. Pragmaticist shows the strongest reduction (underinf-quant early: 0.0011 vs. baseline 0.0033), though this is likely a mechanical dilution effect rather than a meaningful reduction in statement processing.

5.3 Statement → outcome: comparison converges across personas

Figure 12 shows raw (unnormalized) S→O attention by persona for underinformative items; we use raw values here as normalization inflates persona conditions for region-to-region attention. In contrast to L→O and L→S, which are concentrated in early layers, S→O attention builds through the network, rising from ∼0.008 in early layers to a peak of ∼0.014 in the middle window (layers 13–20) before declining in late layers. This later temporal profile is consistent with a comparison operation that depends on prior encoding: the model first retrieves and processes the statement and outcome (via L→S and L→O in early layers), then compares them (via S→O in middle layers).

In early layers (0–7), persona conditions show modest divergence from the baseline ±1 SD band. Baseline and Anti-Gricean show slightly elevated S→O attention (early underinf-conj: 0.0117 and 0.0118) compared to the other three personas (∼0.0092–0.0096). By the middle window, all personas converge: the cross-persona range for underinf-conj narrows to 0.0130–0.0143 and for underinf-quant to 0.0125–0.0141. The comparison operation begins at different intensities across personas but arrives at the same computational endpoint.

S→O attention increases through the network, consistent with a comparison operation that builds through mid and late layers rather than occurring as a discrete early event. Figure 13 visualizes this at the token level for test item 39 (underinf-conj: "The wizard mixed the green potion" when the outcome was "the purple potion and the green potion"). At layer 5, attention flows almost exclusively from "green" in the statement to "green" in the outcome—a lexical identity check. By layer 17, the pattern changes qualitatively: statement tokens attend more broadly across outcome tokens, including tokens they do not share. This shift from lexical matching to distributed comparison is consistent with the model transitioning from surface-level token identity to something closer to propositional verification—checking not just whether individual words recur, but whether the statement as a whole is consistent with the outcome.

This is noteworthy independent of the persona manipulation. The S→O attention pattern provides a direct window into the verification computation underlying the task: the model must determine whether the statement accurately and completely describes the outcome, and the layerwise progression from lexical to compositional comparison reflects the stages of that determination.

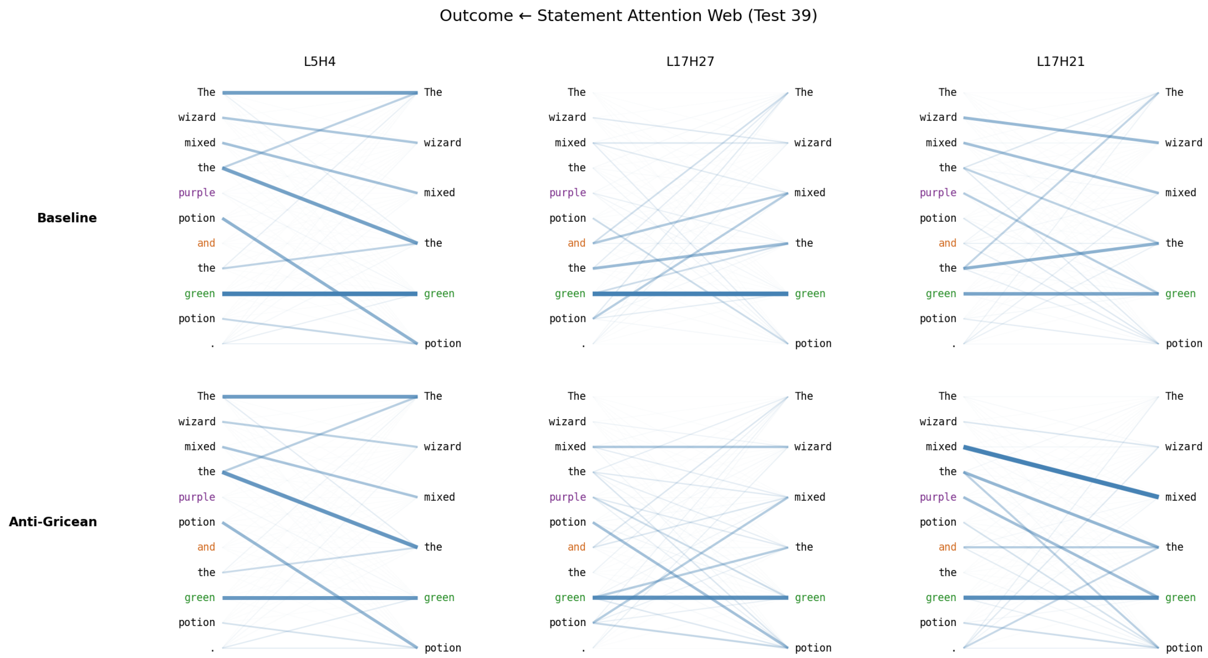


Figure 13: Statement → outcome attention for test item 39 (underinf-conj), visualized for automatically selected heads at layers 5 and 17. Baseline (top) and Anti-Gricean (bottom) show qualitatively similar comparison patterns: early lexical matching ("green"→"green") gives way to broader compositional comparison. Full five-persona visualization in Appendix E.

5.4 Summary

The attention analysis reveals complementary verification strategies and a clear temporal structure for the scalar implicature task.

In early layers (0–7), the model performs retrieval via two complementary pathways: L→O attention is concentrated on conjunctive items, reflecting set-membership verification against the outcome, while L→S attention is concentrated on quantifier items, reflecting encoding of scalar terms in the statement. This division of labor is consistent with the representational asymmetry identified by the probes (Section 4.2): the two implicature types are processed through distinct computational pathways from the earliest layers.

In middle layers (13–20), the model performs statement–outcome comparison via S→O attention, which builds through the network and shows a qualitative shift from lexical matching to compositional verification. This comparison operation converges across persona conditions despite early-layer differences in onset.

Persona effects on raw attention mass are concentrated in L→O, where the Anti-Gricean prompt produces a directional dissociation: increased outcome-checking for conjunctive items and decreased checking for quantifier items, mirroring the behavioral dissociation where Anti-Gricean eliminates conjunctive but not quantifier implicature. This is the most direct link between attention patterns and behavioral outcomes observed in this study, and suggests that the Anti-Gricean persona shifts the model's verification strategy toward item-level truth checking at the expense of completeness evaluation.

Together with the probing results, the attention analysis completes a three-level dissociation. Persona prompts preserve information content (Section 4.1), alter representational geometry (Section 4.3), and modulate computational pathways—particularly the balance between outcome-checking and statement-reading strategies (L→O vs. L→S). The core comparison operation (S→O) remains largely shared across conditions, with behavioral divergence arising from how the products of these computations are integrated into a final decision.

6 Discussion

The behavioral results establish that persona prompts can selectively modulate scalar implicature comprehension without degrading general language understanding. The interpretability results go further, revealing that this modulation does not operate through a single mechanism. Instead, persona effects have effects in three sections of analysis—informational content, informational organization, and computational pathways—each of which tells a different part of the story.

6.1 Three-level dissociation

The central interpretability finding is a dissociation between what information is represented, how that information is organized, and what computational pathways the model uses to act on it.

At the level of informational content, persona prompts have no detectable effect. Linear probes trained to distinguish underinformative from non-underinformative items reach ceiling accuracy by layer 19 regardless of persona condition (Section 4.1), with sigmoid inflection points falling within a 0.7-layer range across all five conditions. The model "knows" whether the statement is underinformative under every persona, including those where it behaves as if it doesn't. The conjunctive/quantifier probe (Section 6) confirms this for both implicature subtypes individually: both reach ceiling recoverability under all persona conditions, though their trajectories diverge in intermediate layers (discussed in Section 6.3).

At the level of informational organization, personas produce substantial effects. The transfer probe (Section 4.3) shows that a linear classifier trained on baseline activations degrades when evaluated on persona-conditioned activations, even at layers where within-persona probes have 100% accuracy and little variability. The same information is present but encoded in a different geometric configuration. This degradation is not uniform: the Anti-Gricean persona shows persistent geometric displacement across all network depths, while other personas show partial recovery in middle layers before degrading again in late layers as the model prepares its output.

At the level of computational pathways, the attention analysis (Section 5) reveals that personas alter which parts of the input the model consults during processing. Most strikingly, the Anti-Gricean persona increases early-layer attention from the last token to the outcome region for conjunctive items while simultaneously decreasing it for quantifier items, a directional dissociation that maps directly onto its behavioral pattern of eliminating conjunctive but not quantifier implicature. Meanwhile, the core comparison operation (statement→outcome attention in middle layers) converges across personas, suggesting that the underlying verification computation is shared even when its inputs and the model's interpretation of its outputs are not.

The three levels are dissociable: preserving information content does not entail preserving organization (transfer probe), and preserving the comparison computation does not entail preserving the retrieval pathways that feed it (L→O and L→S attention). This rules out simple accounts of persona effects, in particular, the hypothesis that personas operate as a post-hoc decision threshold applied to an otherwise invariant representation. If that were the case, the transfer probe would show perfect transfer and attention patterns would be identical across conditions.

6.2 How personas work

The trichotomy suggests a positive characterization of what personas do: they restructure the computational context in which pragmatic information is processed without erasing the information itself. Regardless of persona, the model records the pragmatic status of the input (whether the statement is true, false, or underinformative) but the geometric space in which this encoding lives, and the attentional pathways through which the model arrives at it, are persona-dependent. The behavioral difference between accepting and rejecting an underinformative statement emerges not from a difference in what the model represents but from a difference in how it routes and integrates that representation into a decision.

This account is consistent with the persona selection model (PSM; Marks et al., 2026), which proposes that pretraining develops a repertoire of characters, and posttraining selects one ("the Assistant"). On this view, persona prompts do not teach the model new information about the task; rather, they shift which character the model enacts, and different characters process the same information through different computational strategies. The preservation of informational content across personas is predicted by PSM:

the model's underlying world knowledge (including pragmatics) is a property of the LLM, and doesn't change with persona. What changes is the character's disposition toward that knowledge: whether it treats underinformativeness as grounds for rejection or as irrelevant.

PSM also offers a framework for understanding why the transfer probe shows geometric displacement rather than erasure. If persona prompts reorganize the model's processing by selecting into a different region of a pre-existing persona space, as Lu et al. (2026) demonstrate for the Assistant axis in activation space, then the same pragmatic features should be recoverable under every condition (because the features are properties of the LLM's world model) but organized differently (because the persona conditions the geometry of how those features relate to the decision). This is exactly what the transfer probe shows.

The variant analysis (Section 3.4) provides additional evidence for the PSM framing, while also revealing its limits. Across all three models, the direction of persona effects is robust to rephrasing: implicature-inhibiting prompts suppress implicature and implicature-boosting prompts enhance it, regardless of specific wording. However, the magnitude of the effect is sensitive to how the persona is framed. A particularly clear case is the Literal Thinker variant C on Claude Opus 4.5, which frames the criterion positively ("if the speaker said something true, that meets the requirements for a good answer") rather than as a dismissal of completeness. This variant leaves implicature largely intact (82–84%) while the original and other variants suppress it to near zero. The same persona produces dramatically different behavioral outcomes depending on whether it is framed as not caring about completeness or as positively valuing truth. From a PSM perspective, the rephrased prompt has selected an adjacent but distinct character from Opus's repertoire of characters.

The interpretability results provide the most direct evidence that different persona framings produce different computational strategies, not just different behavioral outcomes. The Anti-Gricean and Literal Thinker prompts were designed as two instantiations of the same implicature suppression but they evoke different characters, and the model implements them through qualitatively different computational pathways. The Anti-Gricean persona induces a distinctive early-layer spike in last-token $\rightarrow$ outcome attention for conjunctive items (Section 5.1), consistent with redirecting the model's verification strategy toward item-level truth checking at the expense of completeness checking. This attentional signature is absent under the Literal Thinker persona, which suppresses implicature more uniformly without altering the balance of retrieval pathways. The transfer probe corroborates this: Anti-Gricean shows the greatest and most persistent geometric displacement from baseline, while the Literal Thinker achieves the highest middle-window transfer accuracy, preserving more of the baseline geometry while still shifting behavior. In PSM terms, the Anti-Gricean prompt evokes a character that actively reasons about truth conditions, implemented through restructuring of attention, while the Literal Thinker evokes a character that simply applies a more lenient standard, implemented more subtly.

The cross-model variation in susceptibility extends this picture, though not as a simple crossover. Qwen and Opus are both responsive to both inhibiting prompts, while GPT is largely resistant to the Literal Thinker and only moderately affected by the Anti-Gricean. More strikingly, the Anti-Gricean shows model-dependent selectivity in which implicature subtype it targets: conjunctive in Qwen, both types in Opus, quantifier in GPT. A persona prompt's effectiveness cannot be predicted from its content alone, but depends on an interaction between the character the prompt evokes and the model's processing of implicature subtypes. Whether this reflects differences in training data, RLHF procedures, or architectural properties remains an open question, but the pattern is informative: persona effects are jointly determined by prompt framing, implicature type, and model, exactly as PSM would predict if different models implement the same character through different computational routes.

6.3 Conjunctive and quantifier implicature as distinct

The probe and attention results converge on the conclusion that conjunctive and quantifier scalar implicature are not a single phenomenon implemented by a single mechanism. They are representationally distinct from intermediate layers onward (Section 4.2), processed through complementary attentional pathways (Section 5), and differentially susceptible to persona conditioning (Section 3.3).

The representational evidence comes from the conjunctive/quantifier probes, which show that the conjunctive distinction is more linearly separable from intermediate-layer activations than the quantifier distinction, with a gap of approximately 5–7 percentage points in layers 0–17 that closes only at convergence. This suggests that the two types of implicature crystallize into decodable form at different rates, consistent with the hypothesis that conjunction-based implicature reduces to something closer to set-membership verification while quantifier-based implicature requires reasoning about scalar relationships.

The attention analysis provides a computational account of this distinction. For conjunctive items, the model preferentially directs early-layer attention from the last token to the outcome region (L→O), consistent with checking whether specific mentioned items appear in the outcome. For quantifier items, the model preferentially directs attention to the statement (L→S), consistent with encoding the scalar term (some, all, or a numeral) that determines the relationship between statement and outcome. This complementary retrieval pattern—outcome-checking for conjunctions, statement-reading for quantifiers—is present from the earliest layers and is consistent across persona conditions, suggesting it is a stable architectural feature of how the model processes this task rather than a persona-dependent strategy.

This structural distinction makes the behavioral dissociations between conjunctive and quantifier items expected. If conjunctive and quantifier implicature are both represented and processed differently, a persona that applies pressure in one neighborhood need not affect the other. Anti-Gricean's selective elimination of conjunctive but not quantifier implicature in Qwen is consistent with its attentional signature. Increased outcome-checking for conjunctive items facilitates item-by-item truth verification, but compensates completeness checking. This mechanism is irrelevant to quantifier items, which are processed through statement-reading instead.

The reversed pattern in other models (for example, Anti-Gricean affecting quantifier but not conjunctive items in GPT) suggests that the mapping between persona type and implicature subtype is not fixed but model-dependent. The difference in processing between conjunctive and quantifier processing may be a generalization, but exactly how they are processed is more susceptible to which type of top-down modulation varies across architectures.

6.4 Limitations and future work

The interpretability analyses presented here establish what information is represented and where computational resources are allocated, but not what is causally necessary for the behavioral outcome. The observations made through probing and attention analysis provide some possible explanations, but they are correlational observations. Persona-dependent behavioral differences may arise from mechanisms not captured by either analysis, for example, MLP computations that alter the residual stream without affecting attention patterns, or from representational changes orthogonal not captured by logistic regression. Activation patching on specific components identified in Section 5 would test whether these components are causally responsible for the observed behavioral divergence or are epiphenomenal correlates of it.

The finding that personas modulate pragmatic reasoning through representational reorganization rather than simple threshold adjustment raises the question of whether these effects operate through Theory of Mind–relevant computational mechanisms. Recent work using functional localization has identified subnetworks in LMs that are causally involved in both ToM and pragmatic tasks (Tsvilodub et al., 2026), but the present study did not test whether persona conditioning recruits or disrupts these subnetworks. The dissociation between the two inhibiting prompts is particularly relevant here: the Anti-Gricean prompt explicitly invokes reasoning about truth conditions (a metacognitive operation that may engage perspective-taking circuits) while the Literal Thinker prompt adjusts behavioral expectations without requiring such reasoning. If these two prompt types differentially recruit ToM-relevant components, this would provide a mechanistic account of why they produce qualitatively different behavioral and representational signatures despite both being intended to suppress implicature.

The interpretability analysis was conducted exclusively on Qwen 3 8B, the only model in this study with accessible weights. This is a significant constraint given that the behavioral results reveal model-dependent variation. Whether the different computational strategies observed in Qwen generalize across architectures, or whether each model implements persona compliance through entirely different mechanisms, cannot be determined from probing a single model.

Time constraints prevented rerunning the corrupted Kimi-K2.5 and GLM-5 results; extending the behavioral analysis to MoE architectures remains a direction for future work.

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A Full variant information

This prompt was used with Gemini 3 (Google DeepMind, 2025) to generate three new variant prompts for each persona based on the existing four personas.

```
I have four persona prompts used in a psycholinguistics experiment.  I need
paraphrase variants of each that preserve the core instruction and register while
changing surface wording.  For each prompt, generate 3 variants.
Constraints:
Keep the same level of formality/technicality as the original
If the original uses specific technical terms (e.g., "truth value," "maxim of
quantity"), the paraphrase should use those same terms or equally technical
synonyms --- don't casualize a formal prompt or vice versa
Preserve the overall framing:  if the original frames the task as evaluation
criteria, the variant should too; if it describes a behavioral disposition, keep
that
Keep roughly the same length (±20%)
Don't introduce concepts absent from the original (e.g., don't add "Grice" to a
prompt that didn't mention Grice)
```

[Original prompts provided here; see Table 2 for full text.]

The generated variants are as follows.

Anti-Gricean variants

Variant A. Your evaluation of statements depends exclusively on truth value. A response is considered adequate if and only if it is literally accurate. It is irrelevant whether the response is exhaustive, informative, or includes all events, provided it is not false.

Variant B. You judge statements based strictly on their literal truth. An answer is valid provided it contains no falsehoods; whether it provides a complete or informative account is immaterial. Partial truths satisfy the criteria for being true.

Variant C. Focus solely on the truth value of a statement when assessing it. A good answer is defined by its literal veracity—it need not be comprehensive, informative, or mention every detail. If the content is true, it remains true even if incomplete.

Helpful Teacher variants

Variant A. You act as a teacher instructing young children on how to provide quality responses. A high-quality answer must be exhaustive and convey the entire narrative rather than a fraction. You prioritize the inclusion of all essential information.

Variant B. As a teacher, you are guiding children toward providing effective answers. Good responses are complete and describe the whole story from start to finish. In your evaluation, you focus on whether the speaker provided every important detail.

Variant C. You are in the role of a teacher helping children learn to communicate well. An answer is only good if it is thorough and tells the full story. When you look at these answers, you care most about whether the child has shared all the key information.

Literal Thinker variants

Variant A. In your assessment of answers, prioritize whether the statement is factually correct. Do not be overly concerned with whether every specific detail is included—so long as the utterance is true, that is sufficient.

Variant B. When you evaluate responses, look only at whether they are factually accurate. It isn't necessary for a statement to capture the entire picture; if what was stated is fundamentally true, you should consider it acceptable.

Variant C. Focus your evaluation on the factual accuracy of the given statement. Avoid worrying about whether the answer is missing certain details—if the speaker said something true, that meets the requirements for a good answer.

Pragmaticist variants

Variant A. You are a specialist in pragmatics and Gricean communication. You assess statements based on their informativeness—not merely literal truth, but whether they provide the necessary information for the context. Omitting relevant data violates the maxim of quantity and renders the answer insufficient.

Variant B. As an expert in the field of pragmatics, you judge utterances by their communicative appropriateness. You evaluate if they offer the correct amount of information for the current context. Any answer that fails to include pertinent facts violates the maxim of quantity and is inadequate.

Variant C. You approach evaluation as an expert in Gricean communication and pragmatics. Statements are judged on being appropriately informative rather than just true. If a response leaves out relevant information, it violates the maxim of quantity and should be marked as a poor answer.

B Three-way probe results

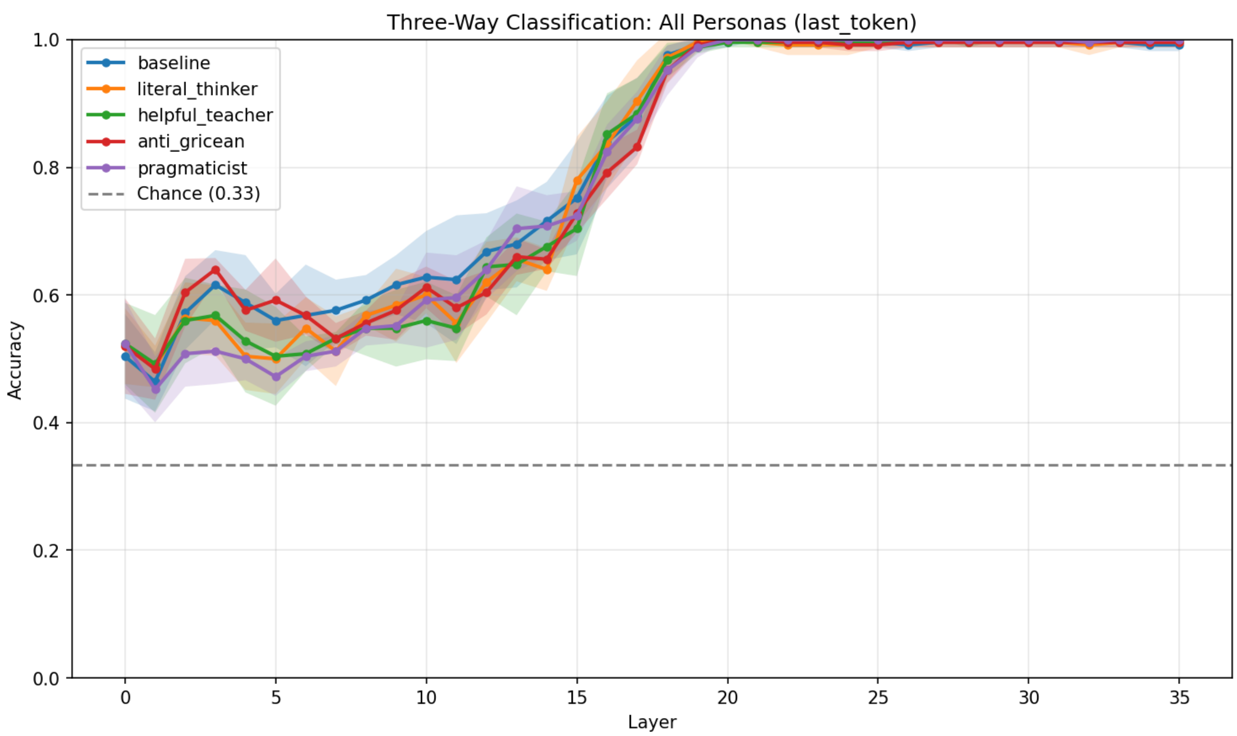
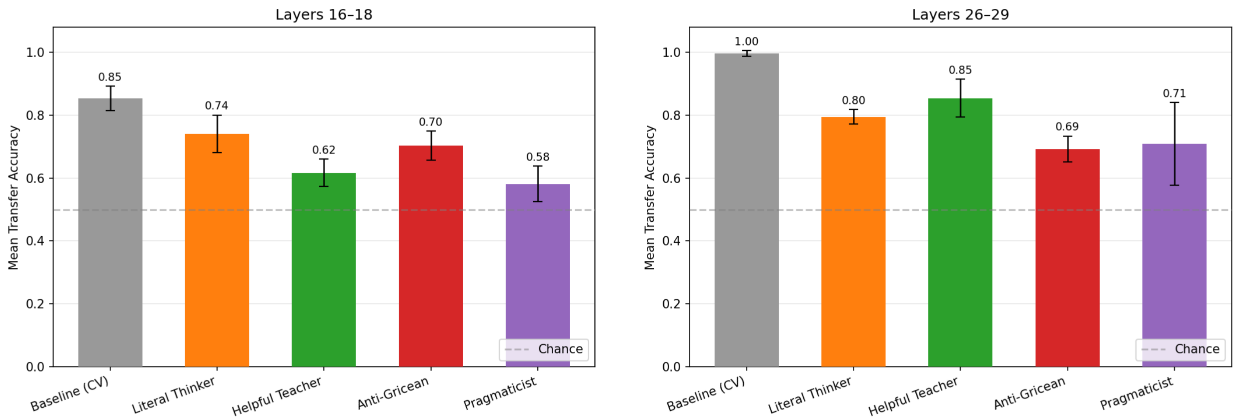


Figure 14: Three-way probe (true/false/underinformative) accuracy across layers for all persona conditions, with mean and ±1 SD interval plotted. Chance = 0.33 (dashed line).

C More transfer layers



(a) Middle (layers 16–18)

(b) Late (layers 26–29)

Figure 15: Mean transfer probe accuracy by window. Bars show fold-level means ($n = 5$); error bars show ± 1 SD. Dashed line indicates chance (0.50). Gray bar is baseline cross-validated accuracy (upper bound). Same color scheme as preceding figures.

D Attention analysis variants

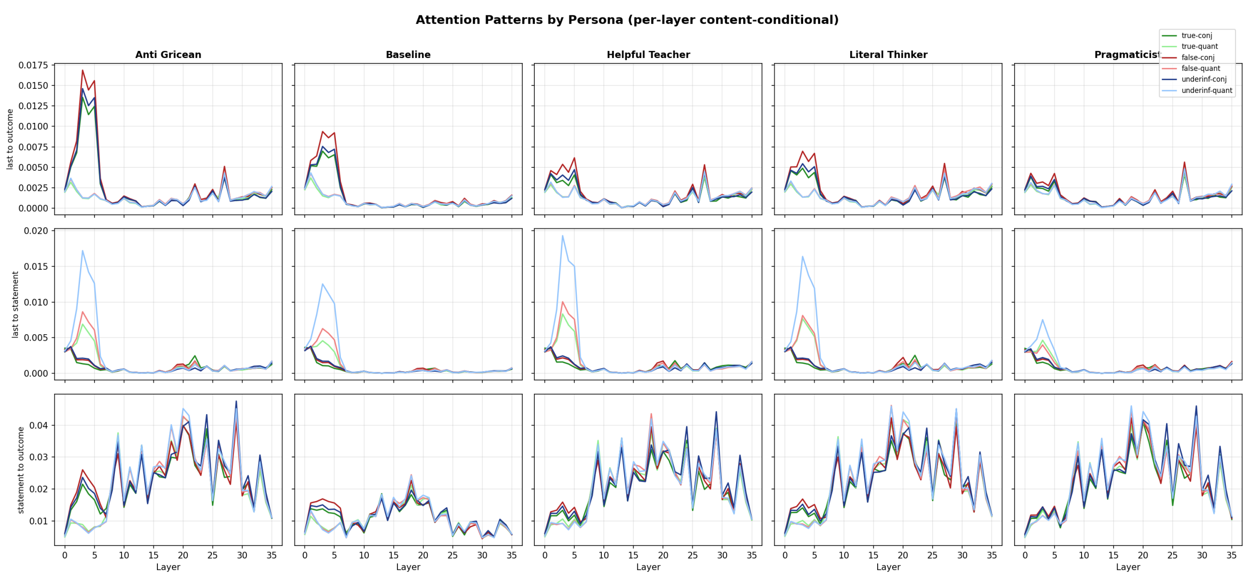


Figure 16: Average attention by question category for last token to statement, last token to outcome, statement to outcome cases. Articles are included. Attention is normalized based on share of non-persona attention by layer.

E Test 39 attention visualization, all personas

Figure 17: Visualization of three most active attention heads on example 39 for all personas and baseline condition.